

CM 7028 Digital Media Production



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FUNDAMENTALS

DIGITAL DATA 21

Digital Data

Bits are units of data that can only have one of two values.

A byte is eight bits.

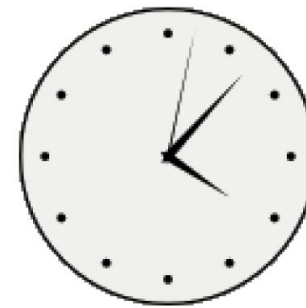
Groups of bits can be interpreted as numbers to base 2, but can also be treated as characters, colours, etc.

Analogue data must be converted to digital form before it can be manipulated by a computer program.

Digitization comprises two operations: sampling and quantization.

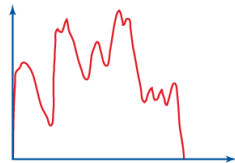
Sampling (graphics), converting continuous colors into discrete color components

Quantization is the procedure of constraining something from a continuous set of values (such as the real numbers) to a relatively small discrete set (such as the integers).

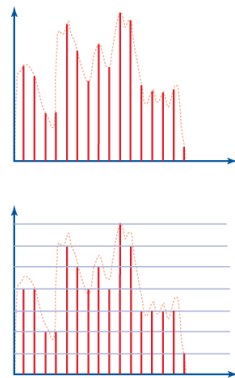


16:07:02

Analogue and digital representations



An analogue signal



Sampling and quantization

The sampling rate is the number of samples in a fixed amount of time or space.

The quantization levels are the set of values to which a signal is quantized.

Digital Media

Website

SEO and SEM

Key internet marketing strategies

- Search engine optimisation (SEO)
 - no cost per click
- Search engine marketing (SEM)
 - cost per click

Slides adapted from Siang Tay's, University of South
Australia / University of Adelaide

Why is internet marketing important?

It gives companies the ability to:

- Drive high quality customers to their website
- Increase sales leads from customers looking for their products and services
- Build their brand online by communicating marketing messages to their target audience
- Increase their profile against their competitors
- Target a global audience via international search engines
- Be accountable with ROI tracking (SEM)

Slides adapted from Siang Tay's, University of South
Australia / University of Adelaide

Search engine optimisation

What is SEO?

SEO is the act of modifying a website to increase its ranking in *organic* (vs *paid*), crawler-based listings of search engines

How do organic search listings work?

- A *spider* or *crawler* which is a component of a SE gathers listings by automatically "crawling" the web
- The spider follows links to web pages, makes copies of the pages and stores them in the SE's index
- Based on this data, the SE then *indexes* the pages and *ranks* the websites
- Major SEs that index pages using spiders: Google, Altavista, Msn, Aol, lycos

"Spiders read only text, nothing else"

Slides adapted from Siang Tay's, University of South Australia / University of Adelaide

Search engine optimisation/cont'd

SEO strategies/techniques

1. Domain name strategies

- domain names are traffic magnets
 - > 63 million domain names registered
 - > 4.7 million domain names registered 1st quar 2004
- choose a domain name that will increase your search engine ranking. How?
 - simple, short, no hyphens, no numbers
 - use keywords, common words, advertising terms, product names
 - choose a keyword that is important for your business

Slides adapted from Siang Tay's, University of South Australia / University of Adelaide

Search engine optimisation/cont'd

2. Linking strategies

- the more inbound links the higher the SE ranking
- if the site linking to you is already indexed, spiders will also receive your site
- quality of inbound links is critical
- how to increase links: a) good content b) good outbound links c) target a list of sites from which you can request inbound links
- links for the sake of links can damage your search rankings

“Link relevancy is critical in getting your site indexed by search engines”

“A small number of inbound links from high-quality, relevant sites is more valuable than many links from low-traffic, irrelevant sites.”

Slides adapted from Siang Tay's, University of South Australia / University of Adelaide

Search engine optimisation/cont'd

3. Keywords

- **important** in optimising rankings
- keywords are words that appear the most in a page
- the spider chooses the appropriate keywords for each page, then sends them back to its SE
- your web site will then be indexed based on your keywords
- can be key phrases or a single keyword
- do not use common words eg 'the' 'and' 'of': spiders ignore them
- write keyword-rich text
- balance keyword-rich and readability
- always have text in your page: at least 100 words

Slides adapted from Siang Tay's, University of South Australia / University of Adelaide

Search engine optimisation/cont'd

4. Title tags

- **important** in optimising rankings
- the first thing that a search engine displays on a search return
- must keywords in title to be ranked no. 1
- should have the exact *keyword* you use for the page
- every single web page must have its own title tag
- you can use up to 65 characters eg Ebay
<http://www.ebay.com/> (*title is littered with keywords*)

5. Meta description tags

- the next important
- displayed below the title in search results
- use dynamic, promotional language
- use keywords

Slides adapted from Siang Tay's, University of South
Australia / University of Adelaide

Search engine optimisation/cont'd

• Example of Meta description tags

```
1 <html><head>
2
3
4 <title>Asst. Prof. Dr. Chutisant Kerdvibulvech</title>
5 <link rel="stylesheet" type="text/css" href="index_files/style.css">
6 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
7 <meta name="description" content="Chutisant Kerdvibulvech">
8 <meta name="keywords" content="PhD, NIDA, National Institute of Development Administration,
9 Thai academic, scholar, columnist, Thailand, Computer Engineering">
10
11 </head><body>
12
13 <table id="outer_frame" align="center" cellpadding="0" cellspacing="0">
```

Slides adapted from Siang Tay's, University of South
Australia / University of Adelaide

Search engine optimisation/cont'd

Meta keywords tags

- no longer carry weight with major SEs
- a myth that meta keywords alone affect rankings

6. Alt tags

- include keywords in your alt tags

7. Submit your website to SEs for indexing

- submit your site to search engine directories, directory sites and portal sites
- indexing takes 1 wk to 3 months

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Search engine optimisation/cont'd

Major SEs to submit to:

- Altavista <http://addurl.altavista.com/sites/addurl/newurl>
(Search engines that also use this database - Looksmart)
- The Open Directory Project (DMOZ) <http://dmoz.org/add.html>
(Search engines that also use this database - Ask Jeeves, Lycos, Netscape)
- Fast http://www.alltheweb.com/add_url.php
(Search engines that also use this database - Lycos)
- Google <http://www.google.com/addurl.html>
(Search engines that also use this database - Yahoo)
- Inktomi <http://submitit.bcentral.com/msnsubmit.htm>
(Search engines that also use this database - AOL, Excite, MSN, Overture)
- MSN <http://submitit.bcentral.com/msnsubmit.htm>
- Yahoo <http://docs.yahoo.com/info/suggest>

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Search engine optimisation/cont'd

SEO - what is NOT recommended

Flash and shockwave - spiders do not pick up these files

Image only sites - spiders do not pick up images

Image maps - spiders cannot read image maps.

In HTML and XHTML, an image map is a list of coordinates relating to a specific image, created in order to hyperlink areas of the image to different destinations (as opposed to a normal image link, in which the entire area of the image links to a single destination). For example, a map of the world may have each country hyperlinked to further information about that country.

Do not use them on your home page or critical pages.

Password protected pages — spiders cannot enter password protected pages

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Search engine optimisation/cont'd

PDF files - can be problematic for spiders. Although some search engines can index them, the pages must be interpreted into HTML and can lose much of their content.

- place PDFs lower down in your site

Drop down menus — spiders cannot read them

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Search engine marketing

What is search engine marketing (SEM)?

SEM is the act of marketing a website via search engines by purchasing paid listings

What are paid listings?

These are listings that search engines sell to advertisers, usually through *paid placement* or *paid inclusion* programs. In contrast, organic listings are not sold.

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Search engine marketing/cont'd

Paid listings:

1. Paid inclusions

- Advertising programs where pages are guaranteed to be included in a search engine's index in exchange for payment
- no guaranteed ranking
- payment made on a *Cost Per Click (CPC)* basis
- eg Looksmart: a directory that lists pages and sites, not based on position but based on relevance. Advertisers pay to be included in the directory on a CPC basis or per-url fee basis with no guarantee of specific placement

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Search engine marketing/cont'd

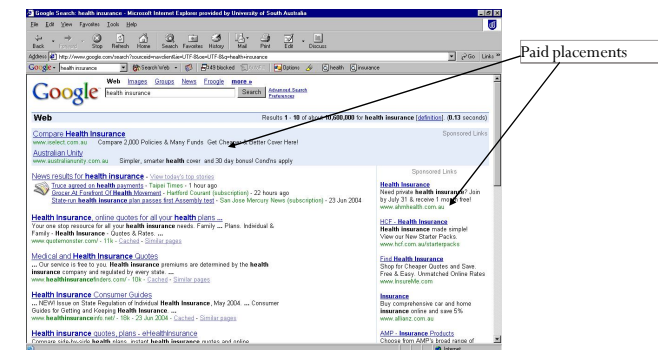
2. Paid placements

- Advertising programs where listings are guaranteed to appear in organic listings
- the higher the fee, the higher the ranking
- eg **sponsored links** and **Google's Adwords**
- can be purchased from a portal or a search network
- search networks are often set up in an auction environment where keywords and phrases are associated with a cost-per-click (CPC) fee.
- Overture and Google are the largest networks

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Search engine marketing/cont'd

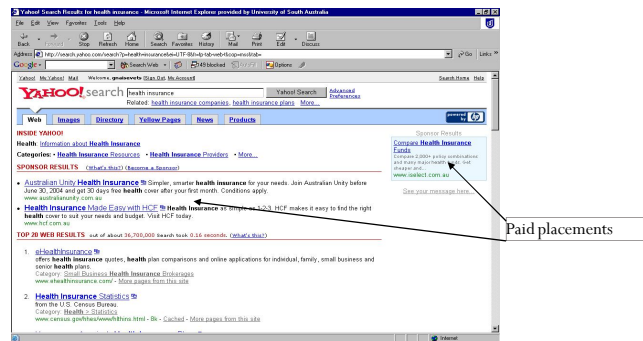
Paid placements - Google



Slides adapted from Siang Tay's, University of South Australia / University of Adelaide

Search engine marketing/cont'd

Paid placements - Yahoo



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Thank you

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